

Switching Perspectives: A Preliminary Study on VR Empathy Training for Distracted Cycling

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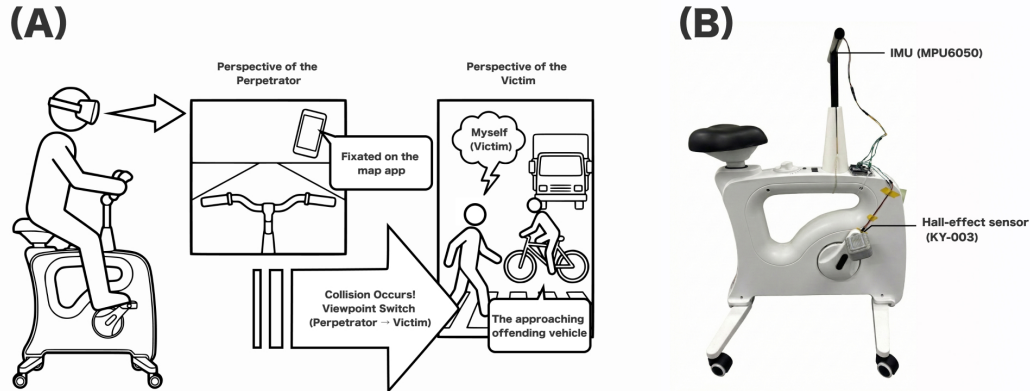


Figure 1: System Design_ (A) Perspective-Switching Logic and (B) Hardware Setup

Abstract

Smartphone usage while cycling creates significant traffic hazards, yet traditional safety education often fails to help youths internalize these risks due to a lack of embodied experience. In this work, we introduce a VR-based educational system designed to deter distracted cycling through a novel perspective-switching mechanism. The system simulates a collision caused by smartphone usage and immediately forces a viewpoint shift from the cyclist (perpetrator) to the pedestrian (victim), compelling users to experience the accident's consequences from the injured party's perspective. We conducted a pilot study (N=12) to evaluate user experience and changes in risk perception. Preliminary findings suggest that this dual-perspective approach may help reduce optimism bias and support empathy, with participants reporting a heightened sense of danger when placed in the victim's role. We discuss the design implications of using embodied perspective-taking to promote altruistic behavioral change in traffic safety education.

CCS Concepts

• Applied computing → Interactive learning environments.

Keywords

Virtual Reality, Distracted Cycling, Perspective Switching, Embodiment, Traffic Safety, Pilot Study

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1 Introduction

Smartphones have become indispensable tools in daily life, yet their usage while moving—specifically distracted cycling—has emerged as a prominent social problem and a significant threat to traffic safety. Despite the popularity of bicycles as a convenient mode of transport for young people, accidents resulting from smartphone use while riding are increasing. In Japan, traffic safety reports and police statistics indicate that approximately 80% of distracted cycling-related accidents involve individuals under the age of 29 [9]. While stricter regulations, such as the 2024 amendment to Japan's Road Traffic Act, have been implemented to penalize this behavior, legal enforcement alone is often insufficient to induce sustained personal behavioral change. Traditional safety education methods, such as fear appeals or passive lectures [4], frequently fail to help users internalize these risks, leading learners to dismiss accidents as irrelevant to themselves. Virtual Reality (VR) technology, with its ability to simulate dangerous scenarios in a safe environment, offers a promising alternative for traffic safety education. VR enables



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users to experience the consequences of unsafe behaviors without physical harm. However, existing VR bicycle simulators often focus heavily on general traffic rules, such as obeying signals or stop signs [6], with limited attention given to the specific mechanics and psychological triggers of distracted cycling. Unlike general riding, using a smartphone while cycling imposes unique cognitive and physical loads—such as reduced situational awareness and unstable steering [2]—which demand specialized training to bridge the gap between knowing the rules and understanding the visceral reality of an accident. In this paper, we explore the design of an immersive VR educational system aimed at preventing smartphone use while cycling by internalizing the risks through experience rather than knowledge alone. Our research addresses the limitations of external controls by fostering intrinsic motivation for safety. To achieve this, we developed a system that allows users to actively use a smartphone while cycling, leading to a simulated accident. The system incorporates a perspective-switching mechanism (Figure 1) that shifts the user's viewpoint from the perpetrator (cyclist) to the victim (pedestrian) immediately after a collision. We examine how this dual-perspective experience may cultivate empathy and reshape risk awareness, offering insights into how VR can be designed to promote altruistic behavioral change in traffic safety.

2 Related Work

This study proposes a VR-based educational system that utilizes perspective-switching to allow users to experientially learn the dangers of smartphone use while cycling. In this section, we outline the theoretical background and technological challenges in constructing this system.

2.1 Limitations of Video-Based Safety Education

Currently, threat appeal—showing shocking accident footage to evoke fear—remains a primary method in traffic safety education. However, research indicates that video viewing alone is often insufficient to induce behavioral change, particularly among younger demographics [7]. This limitation stems from three key factors: a lack of personal relevance, the absence of cognitive load simulation, and a lack of embodiment. First, young people often exhibit an optimism bias, processing accident footage as events that "will not happen to me" [7]. Unidirectional lectures fail to help learners internalize these risks, preventing the development of the "coping appraisal" necessary for behavioral modification [8]. Second, video viewing does not simulate the dual-task nature of distracted cycling. Cycling while using a phone increases mental workload and delays reaction times [3]. Video viewing, being a single-task activity where attention is fully focused on the screen, fails to recreate the divided attention and inattentive blindness inherent in the actual behavior [11]. Finally, traditional media lacks physicality. Chittaro et al. (2014) demonstrated that, unlike passive media, VR promotes embodied cognition, which is highly effective for safety education [1]. To change habitual behaviors—such as reflexively reaching for a phone—knowledge transfer is insufficient. Instead, an active learning system that utilizes "Place Illusion" and "Plausibility Illusion" [11] to allow learners to physically experience failure in a safe environment is essential. Beyond traffic safety, VR perspective-switching has been examined in social psychology as a means of

fostering empathy, such as placing domestic violence offenders in the victim's perspective. The present study applies this mechanism to distracted cycling, targeting the specific cognitive biases of road users [10].

2.2 VR Bicycle Simulators and Interaction Design

In recent years, numerous VR bicycle safety education systems have been proposed. These studies generally focus on teaching traffic rules or improving hazard prediction, utilizing various software and hardware approaches. On the content side, Nagami et al. developed a simulator using Head-Mounted Displays (HMDs) to teach responses to common accident scenarios, such as sudden lane changes [6]. Conversely, Wang et al. proposed a "Choose-your-own-adventure" style app using gaze-based input [14]. While effective for general rules, these systems often lack a focus on the specific dual-task scenario of smartphone manipulation while riding. Regarding hardware, the trade-off between realism and stability remains a challenge. To replicate realistic bicycle behavior, Ullmann et al. developed "BikeVR," which allows the bicycle to tilt physically [13]. Similarly, Matviienko et al. investigated the effects of steering methods on VR motion sickness [5]. However, these studies suggest that realistic tilting can induce motion sickness and instability. For a system designed to test distracted cycling—where the user holds a smartphone with one hand and has limited visual attention—unstable hardware (like tilting bikes) poses a significant risk of falling and hinders the accurate measurement of the distraction itself. Alternatively, Tsuboi et al. utilized a stationary exercise bike combined with VR in a gamified manner-learning system [12]. While their focus was on general riding etiquette rather than high-risk distraction tasks, the stability of the exercise bike is advantageous. Consequently, this study adopts a stable exercise bike framework combined with a custom controller. This approach addresses the limitations of prior work by ensuring user safety while enabling the reproduction of high-risk behaviors, such as one-handed steering and gaze aversion, which are difficult to safely simulate with unstable, realistic hardware.

3 System Design

3.1 Core Features

We developed the application using the Unity game engine and the Meta Quest 3 HMD to simulate the cognitive load of dual-task riding. The system focuses on two core mechanisms: smartphone interaction and a two-stage perspective switching. Unlike passive simulators, our system encourages users to view a virtual smartphone map for navigation while riding. When a collision occurs, the system executes a scripted transition sequence: first, the user remains in the perpetrator's POV for 5 seconds to observe the immediate aftermath. The user is then transported to a cinematic virtual space to experience the event from the victim's (pedestrian) POV, beginning five seconds before impact. This perspective shift allows the user to observe the approaching bicycle and the collision from the victim's standpoint. The mechanism is intended to mitigate optimism bias and support risk awareness by providing a more objective view of the accident's consequences.



Figure 2: The view transition from Perpetrator (Left) to Victim (Right)

3.2 Hardware Implementation

To bridge the gap between physical realism and user safety, we developed a custom controller interface using a stationary exercise bike. For data processing, an Arduino Uno R4 Wi-Fi handled real-time sensor fusion. A Hall-effect sensor (KY-003) detected pedal rotations, which Unity translated into virtual acceleration. An MPU6050 six-axis IMU, mounted on a 3D-printed handlebar joint, transmitted real-time rotational data for steering control. This configuration ensures user safety while enabling the reproduction of high-risk behaviors, such as one-handed steering and gaze aversion.

3.3 Scenario Design

We developed a realistic urban cycling scenario focused on a "Restaurant Finding" task to simulate the cognitive load associated with everyday navigation. The environment was constructed in Unity, featuring a detailed cityscape where buildings create natural blind spots at intersections. Event triggers were embedded at these locations. Unlike a fixed script, collisions were avoidable: participants who maintained situational awareness and steered appropriately could continue the task without incident. However, if the system detected a downward gaze (distraction) as the participant entered a trigger zone, a pedestrian emerged from a blind spot, creating a high-risk situation. This design replicated the visual and cognitive distractions of real-world wayfinding.

4 Evaluation

We conducted a pilot study with university students to examine user experience and potential changes in risk perception related to distracted cycling. This section outlines our participants, procedure, and measures.

4.1 Participants

We recruited 12 university students (6 female, 6 male) aged between 20 and 29. All participants provided informed consent prior to the experiment. The group consisted of a balanced mix of cycling habits, with approximately 50% riding at least once a month and the other 50% riding rarely or never.

4.2 Procedure

The experiment was conducted in a controlled laboratory setting following a three-step procedure. First, participants completed a pre-study survey to assess their demographics, cycling habits, and baseline risk perception. Next, participants donned the VR headset and engaged in the VR simulation. They were instructed to navigate a city while using a virtual smartphone, leading to a simulated collision. Crucially, the system executed the perspective-switching mechanism, forcing participants to experience the accident first as the perpetrator (cyclist) and immediately after as the victim (pedestrian). Finally, participants completed a post-study survey to evaluate changes in their awareness, the impact of the perspective switch, and any symptoms of simulator sickness.

4.3 Measures

To assess user experience and possible impacts, we employed a mixed-methods approach utilizing pre- and post-study surveys. The pre-survey collected demographic information and assessed participants' daily cycling habits and specific smartphone usage patterns (e.g., navigation, messaging) to establish a baseline. It also measured their initial risk perception, including threat appraisal (perceived danger) and optimism bias (belief in one's own immunity to accidents), as well as their coping appraisal (self-efficacy). Following the VR intervention, the post-survey re-evaluated these risk perception metrics to quantify any shifts in awareness. Additionally, the post-survey assessed the user experience, specifically focusing on the emotional impact of the perspective-switching mechanism—comparing the shock felt as a victim versus a perpetrator—and monitored for simulator sickness. All subjective measures were rated on a 5-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree."

5 Preliminary Findings

We analyzed the survey data to evaluate the system's impact on risk perception and user experience. Our findings are organized across three categories: baseline habits, awareness change, and user feedback.

5.1 Baseline Habits and Awareness

The demographic analysis (N=12) revealed a split in cycling habits. While 58% (N=7) stated they never use smartphones while riding, 42% (N=5) admitted to occasional use, primarily for navigation and hands-free calling rather than high-load tasks like gaming. Despite this, participants displayed high self-efficacy, with 75% (N=9) strongly believing they could refrain from usage if they chose to. However, baseline data indicated a latent optimism bias, where several participants expressed uncertainty about whether their personal cycling skills could mitigate accident risks.

5.2 Changes in Risk Perception

We compared pre- and post-experiment scores to measure awareness change. Prior to the VR session, risk perception was ambiguous for some participants. Post-experiment, this ambiguity vanished. Notably, we observed a reduction in optimism bias in this pilot study: while some participants initially believed their skill or caution could prevent accidents, 100% rejected this notion after the simulation. The experience appeared to challenge the ‘it won’t happen to me’ mindset, with all participants in our sample agreeing that distracted cycling can pose a serious risk of injury.

5.3 User Experience and Feedback

All participants reported feeling fear when placed in the victim’s perspective (N=12,100%). Notably, 75% (N=9) indicated that the accident had a stronger impact when experienced as the victim (pedestrian) rather than as the perpetrator (cyclist). These findings suggest that, in this pilot, being ‘hit’ was perceived as more impactful than hitting another person. Overall, 92% (N=11) rated the system as ‘effective’ in the post-study questionnaire. However, a physiological challenge remains: 66% (N=8) reported experiencing simulator sickness, likely due to the sensory conflict during the collision sequence.

6 Discussion, Conclusion, And Future Work

This study explored the potential of a VR simulator featuring perspective switching to influence risk perceptions of smartphone use while cycling. Pilot findings suggest the system may reduce optimism bias, particularly among habitual users who initially believed their riding skills could prevent accidents. By requiring participants to experience the consequences of a collision from the victim’s viewpoint, the system functioned as a simulated failure experience that appeared to challenge unfounded confidence more strongly than passive observation-based learning. The increase in awareness appeared more pronounced among daily cyclists, indicating that this embodied approach may be particularly relevant for addressing normalized unsafe behaviors. Despite the positive educational outcomes, feedback highlighted several areas for improvement. Participants requested enhanced mechanical realism, such as functional braking and complex traffic environments, to better mirror real-world unpredictability. Furthermore, our baseline survey indicated a high prevalence of auditory distractions, such as calling and music listening, which were not covered in the current visual-focused scenario. Future iterations should incorporate these auditory tasks to address a broader range of risk behaviors. Finally, reducing simulator sickness remains a priority to ensure

the tool is viable for widespread training. In conclusion, while further refinement in scenario diversity and comfort is needed, the perspective-switching mechanism offers a promising direction for fostering genuine safety awareness. As a preliminary investigation, this study lacked a control condition (non-switching group) to fully isolate the perspective-switching effect. In addition, the relatively high rate of simulator sickness (57.1%) may have influenced participants’ responses; future iterations will implement visual anchors and more gradual transitions to mitigate this.

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References

- [1] Luca Chittaro. 2014. Designing serious games for safety education: “Learn to brace” versus traditional pictorials. *IEEE Transactions on Visualization and Computer Graphics* 20, 11 (2014), 1576–1586.
- [2] Dick De Waard, Paul Schepers, Wytse Ormel, and Karel Brookhuis. 2010. Mobile phone use while cycling: Incidence and effects on behaviour and safety. *Ergonomics* 53, 1 (2010), 30–42.
- [3] Kazuhiro Kozuka. 2016. Gaze Measurement of a Pedestrian Texting While Walking and a Rider Texting While Riding a Bicycle, and Verification of Dangerousness of Texting. *Fundamentals Review* 10, 2 (2016), 129–136.
- [4] I. Lewis, B. Watson, R. Tay, and K. M. White. 2007. The role of fear appeals in improving driver safety: A review of the effectiveness of fear-arousing (threat) appeals in road safety advertising. *International Journal of Behavioral Consultation and Therapy* 3, 2 (2007), 203–222.
- [5] Andrii Matvienko, Florian Müller, Martin Zickler, Lucas A. Gasche, Julia Abels, Till Steinert, and Max Mühlhäuser. 2022. Reducing virtual reality sickness for cyclists in VR bicycle simulators. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI ’22)*. ACM, New York, NY, USA, Article 187, 14 pages.
- [6] Yutaka Nagami and Fumiya Fuse. 2023. Experiential program with crisis avoidance using VR bicycle simulator / Verification of realism with head-mounted display. In *Proceedings of the 70th Annual Conference of JSSD*, Vol. 70. 508.
- [7] Yukako Nakano. 2017. A Review of the Studies on the Scared Straight Traffic Safety Programs. *Traffic Science (Kotsu Kagaku)* 48, 2 (2017), 40–46.
- [8] Yukako Nakano, Kazuko Okamura, Ritsu Kosuge, and Goro Fujita. 2021. The Effectiveness of Group Discussion in a Threat Appeal -based Road Safety Education for Young Cyclists. *Japanese Journal of Traffic Psychology* 37, 1 (2021), 16–29.
- [9] National Police Agency Traffic Bureau. 2024. Traffic Accident Occurrence Situation in 2024. <https://www.npa.go.jp/publications/statistics/koutsuu/jiko/R06bunseki.pdf>. Retrieved December 21, 2025.
- [10] S. Seinfeld, J. Arroyo-Palacios, G. Iruretagoyena, et al. 2018. Offenders become the victim in virtual reality: impact of changing perspective in domestic violence. *Scientific Reports* 8, Article 2692 (2018).
- [11] Mel Slater. 2009. Place illusion and plausibility illusion. *Philosophical Transactions of the Royal Society B: Biological Sciences* 364, 1535 (2009), 3549–3557.
- [12] H. Tsuboi, S. Toyama, and T. Nakajima. 2018. Enhancing bicycle safety through immersive experiences using virtual reality technologies. In *Augmented Cognition: Intelligent Technologies (AC 2018)*, D. Schmorow and C. Fidopiastis (Eds.). Lecture Notes in Computer Science, Vol. 10915. Springer, Cham, 250–264.
- [13] Daniel Ullmann, Julian Kreimeier, Timo Götzmann, and Holger Kipke. 2020. BikeVR: A virtual reality bicycle simulator towards sustainable urban space and traffic planning. In *Proceedings of Mensch und Computer 2020 (MuC ’20)*. ACM, New York, NY, USA, 511–514.
- [14] W. Wang, K. P. Singh, Y. T. M. Chu, and A. Huber. 2016. Educating bicycle safety and fostering empathy for cyclists with an affordable and game-based VR app. In *Proceedings of the 18th International Conference on Human-Computer Interaction with Mobile Devices and Services Adjunct (MobileHCI ’16)*. ACM, New York, NY, USA, 883–890.